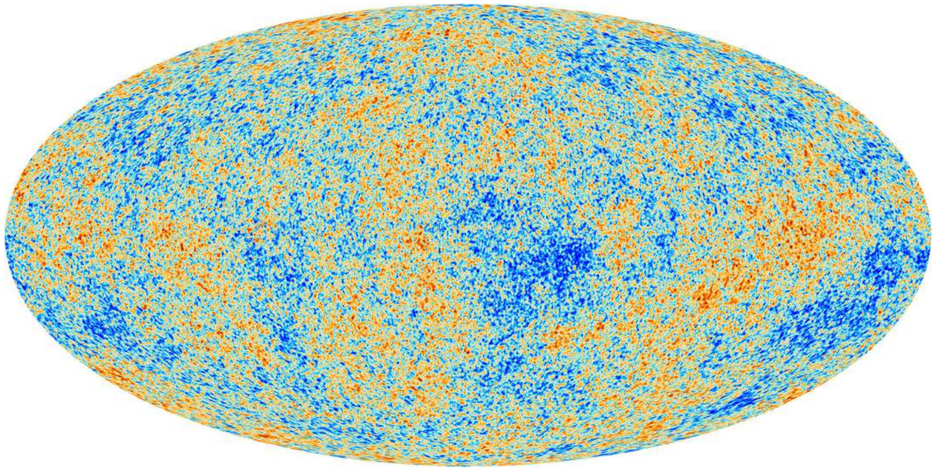

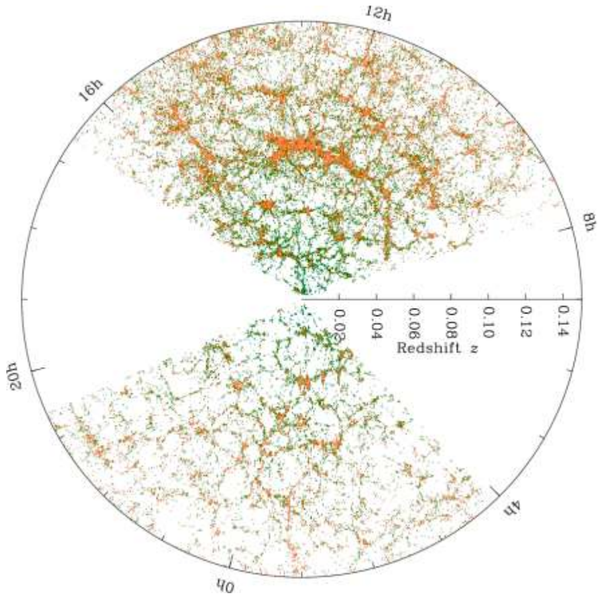
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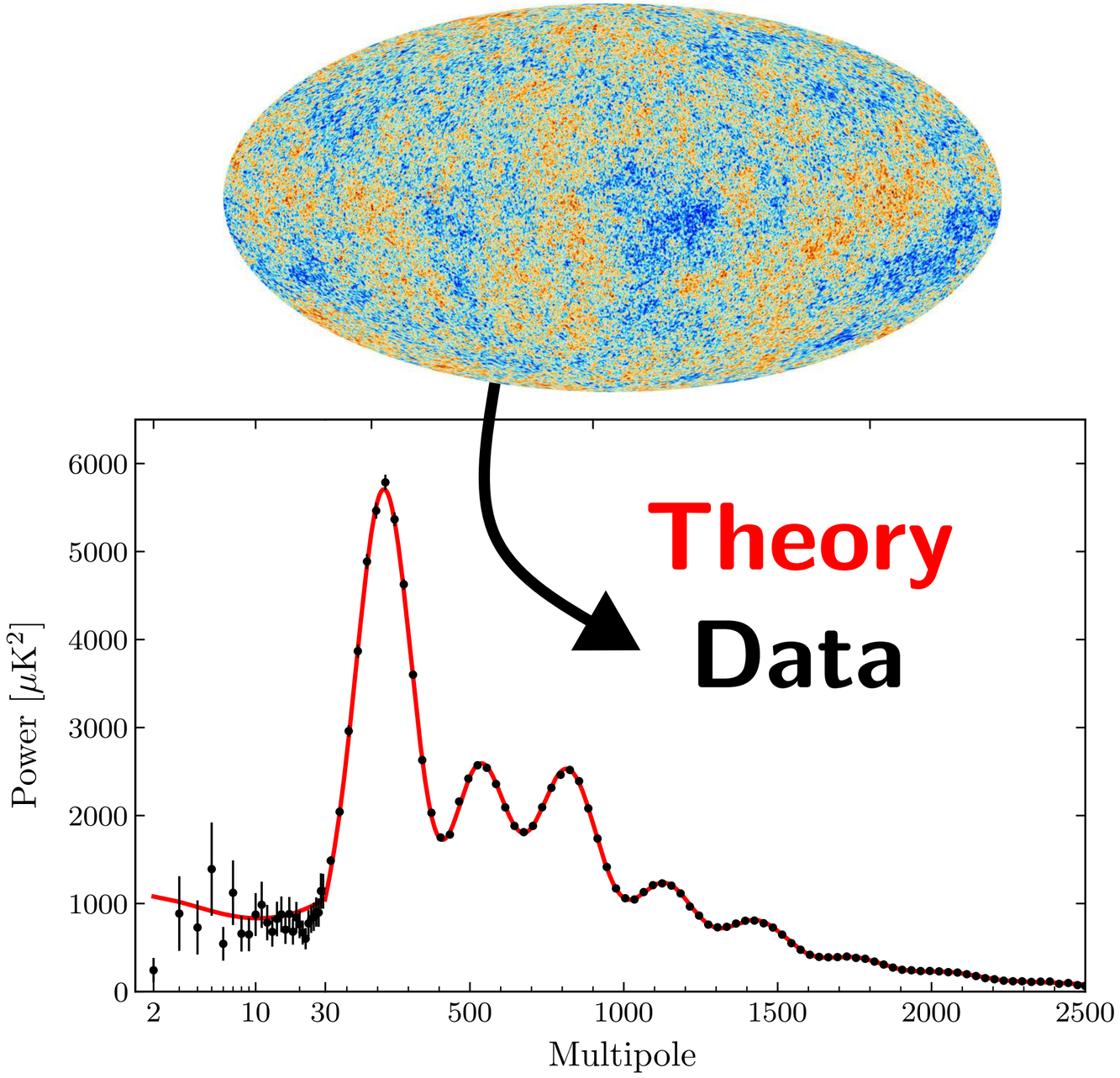
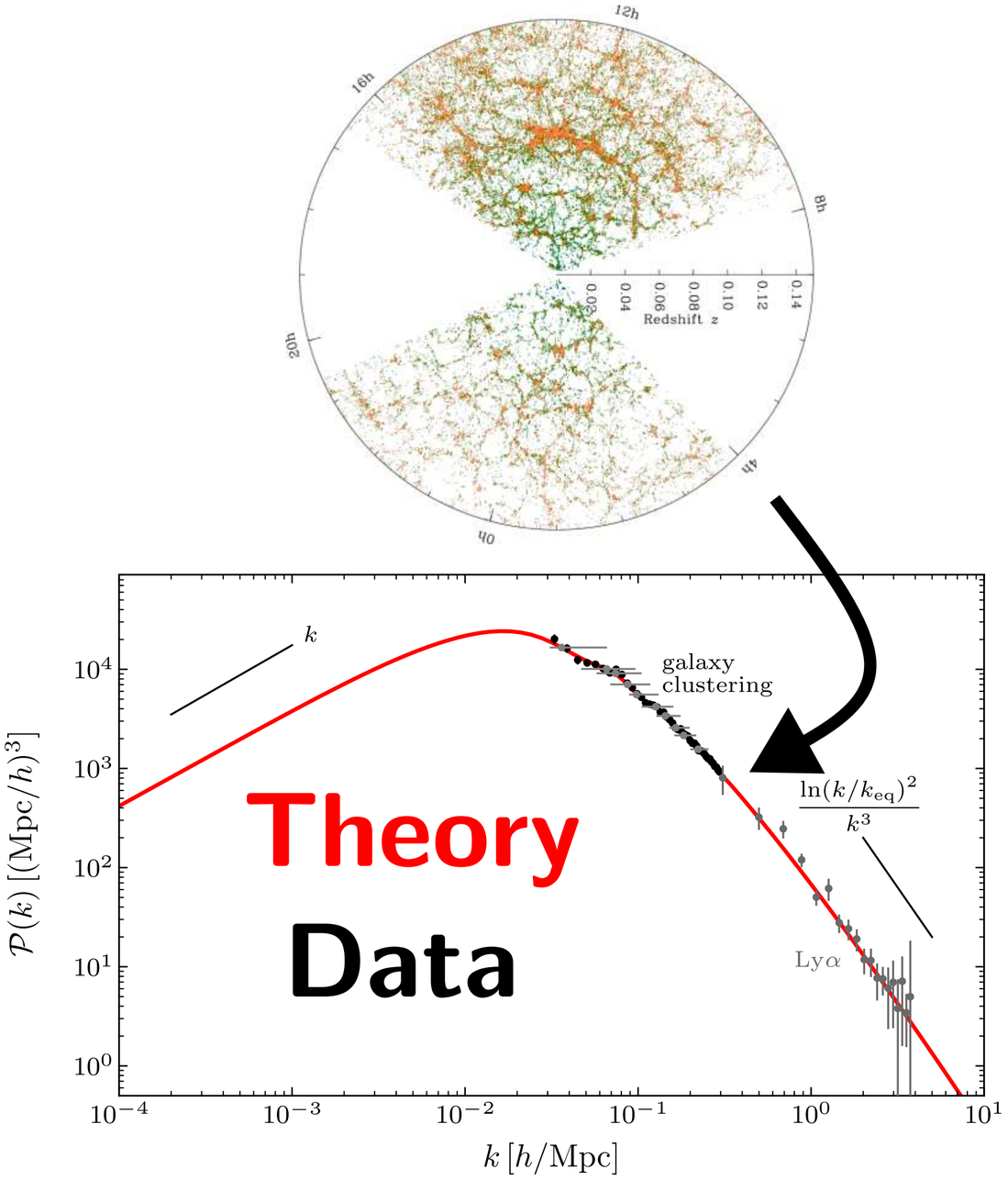
8h
4h
0h
20h
16h
0.03
0.04
0.06
0.08
0.10
0.12
0.14
Redshift z


Theory
Data
Power [μK^2]
Multipole
0
1000
2000
3000
4000
5000
6000
2
10
30
500
1000
1500
2000
2500


12h
16h
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20h
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Redshift z
0.02
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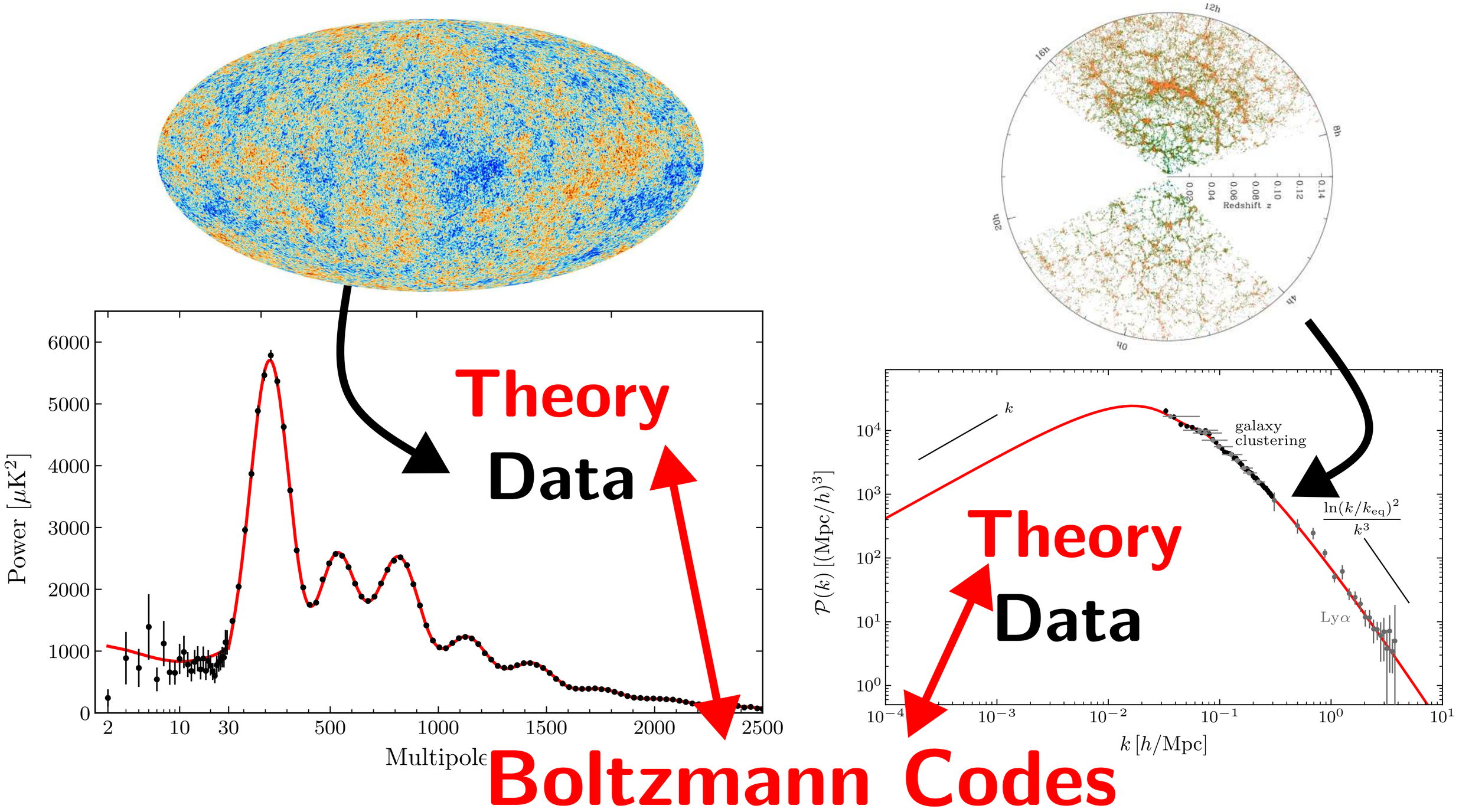
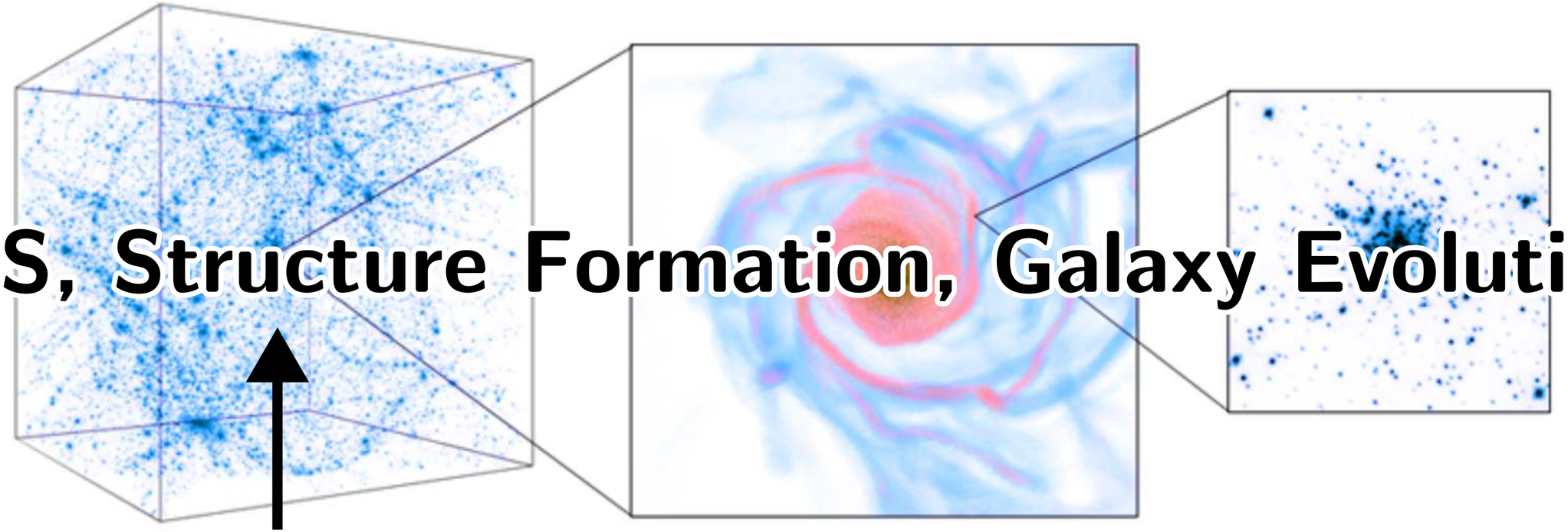
0.12
0.14
Theory
Data
k
galaxy clustering
$\frac{\ln(k/k_{\rm eq})^2}{k^3}$
Lyα
$\mathcal{P}(k)$ [(Mpc/h)^3]
k [h/Mpc]
10^0
10^1
10^2
10^3
10^4
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^0
10^1


Theory
Data
Power [μK^2]
Multipole
Theory
Data
$\mathcal{P}(k)$ [(Mpc/h)^3]
k [h/Mpc]
galaxy clustering
Lyα
$\frac{\ln(k/k_{eq})^2}{k^3}$
Redshift z
Boltzmann Codes



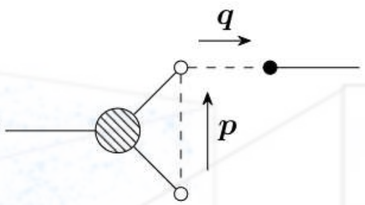
LSS, Structure Formation, Galaxy Evolution

Simulations

Initial Conditions

Boltzmann Codes

$$\langle (v_\parallel^2)_{\boldsymbol{q}} \delta^{(1)}_{\boldsymbol{q}} \rangle = 2 \times$$

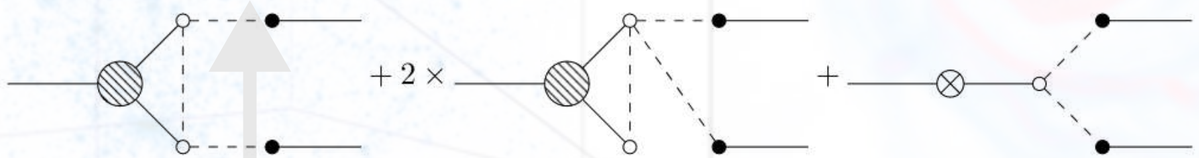


$$= -4\mathcal{H}^2 P(\boldsymbol{q}) \int d^3p\, P(\boldsymbol{p}) \frac{p_\parallel}{p^2} \frac{(\boldsymbol{q}-\boldsymbol{p})_\parallel}{(\boldsymbol{q}-\boldsymbol{p})^2} G_2(-\boldsymbol{p}, \boldsymbol{q})$$

$$= \frac{\mathcal{H}^2}{105} \left(71 + 23 \frac{q_\parallel^2 - q_\perp^2}{q^2} \right) P(\boldsymbol{q}) \varsigma^2(\Lambda) \qquad \text{(A2)}$$

At $n = 2$, there are three diagrams to consider.

$$\langle (v_\parallel^2)_{\boldsymbol{q}_1+\boldsymbol{q}_2} \delta^{(1)}_{\boldsymbol{q}_1} \delta^{(1)}_{\boldsymbol{q}_2} \rangle =$$



(A3)

CLASS-PT: nonlinear perturbation theory extension of the Boltzmann code CLASS

This is a modification of the CLASS code that computes the non-linear power spectra of dark matter and biased tracers in one-loop cosmological perturbation theory, for both Gaussian and non-Gaussian initial conditions.

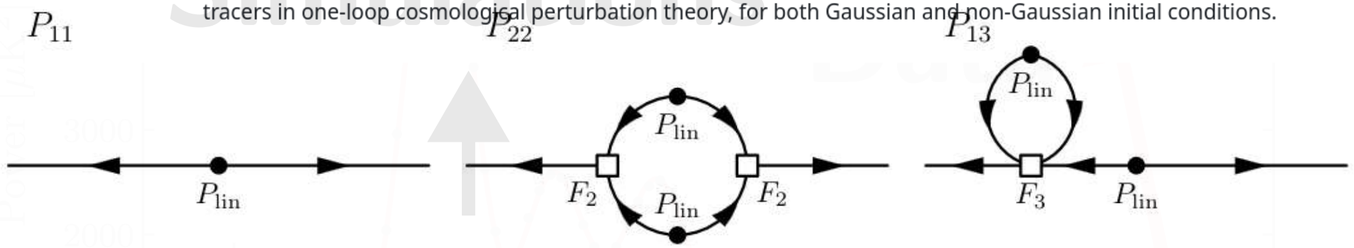


Figure 2.5.: Diagrammatic representation of the one loop matter power spectrum in Eq. (2.66) where $P_{11} = P_{\rm lin}$.

$$\xi_{\rm DM}(r) = \xi_{\rm 1h}(r) + \xi_{\rm 2h}(r)$$

$$\xi_{\rm 1h} = \frac{1}{\rho_M^2} \int {\rm d}m\, m^2 \frac{{\rm d}n(m)}{{\rm d}m} \int {\rm d}^3x\, \lambda_m(\vec{x}) \lambda(\vec{x}+\vec{r})$$

$$\xi_{\rm 2h} = \frac{1}{\rho_M^2} \int {\rm d}m_1 \int {\rm d}m_2 m_1 \frac{{\rm d}n(m_1)}{{\rm d}m_1}\, m_2 \frac{{\rm d}n(m_2)}{{\rm d}m_2} \times \int {\rm d}^3x \int {\rm d}^3y\, \lambda_{m_1}(\vec{x}) \lambda_{m_2}(\vec{y}) \times \xi_{hh}(\vec{x}-\vec{y}+\vec{r} \,|\, m_1, m_2)$$

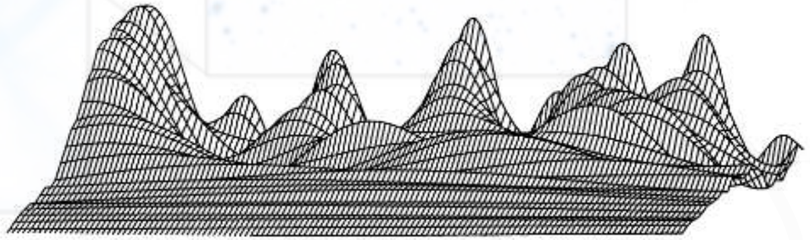


FIG. 1a

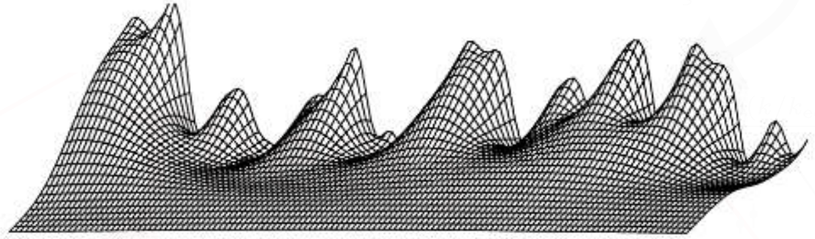


FIG. 1b

FIG. 1.—Topographical maps of one-dimensional Gaussian random density fields $F(r, A)$ as a function of position and resolution for (a) sharp k-space filtering and (b) Gaussian filtering. The spectra are $n = 0$ power laws. The depth dimension is $\log_{10}(R_f)$, with a spacing of 0.06 and depth coverage a factor of 1.5. The horizontal dimension is position r, the range is 1/64 of the box the field was generated on. The range in the rms amplitude is $(32)^{1/2}$.

Perturbative (EFT)

Halo Model

Boltzmann Codes

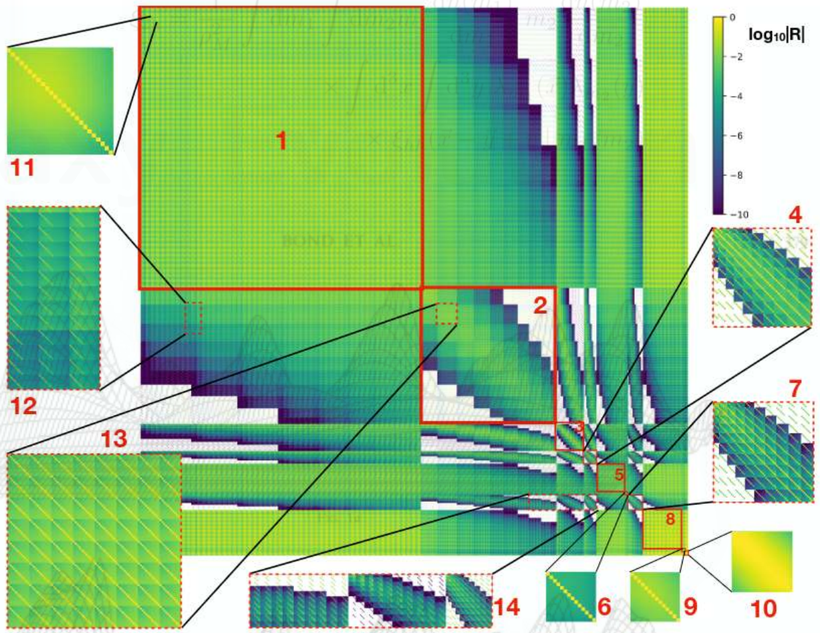


Figure B2. The 10×2pt correlation matrix for LSST Y6 ×SO Y5, plotted as $\log_{10}|\mathbf{R}|$, where $\mathbf{R} = [\mathrm{diag}(\mathbf{C})]^{-1/2}\mathbf{C}[\mathrm{diag}(\mathbf{C})]^{-1/2}$. We have highlighted some parts of the matrix to illustrate the correlation structure: (1) depicts the cosmic shear (κ_g-κ_g) correlation matrix, comprised of 55 tomographic combinations of source bins, each with 25 Fourier ℓ-bins. (11) shows one of the tomographic combinations, and the individual ℓ elements are clearly visible. (2) is the galaxy-galaxy lensing (κ_g-δ_g) tomography correlation with (13) being the galaxy-galaxy combinations of the 2nd lens bin with all the non-overlapping source bins at higher redshifts. (12) zooms into the matrix of a part of the correlation between cosmic shear and galaxy-galaxy lensing. (3) is the clustering (δ_g-δ_g) auto-probe matrix with 10 tomographic bins. (4) corresponds to the galaxy-CMB convergence (δ_g-$\kappa_{\rm CMB}$) combination auto-probe matrix, which is comprised of 10 lens sample redshift bins. (5) is the auto-probe correlation of the κ_g-$\kappa_{\rm CMB}$ part of the data vector, which uses the 10 source sample redshift bins. (6) is the auto-probe correlation of the CMB lensing power spectrum, $\kappa_{\rm CMB}$-$\kappa_{\rm CMB}$. (7) is the auto-probe correlation of the galaxy-tSZ (δ_g-y) combinations, which uses the 10 lens redshift bins. (14) zooms into the correlation matrix of all 10 galaxy-tSZ combinations with the last 5 of galaxy-galaxy lensing power spectra (κ_g-δ_g), all the 10 galaxy clustering (δ_g-δ_g), and all 10 δ_g-y spectra, respectively from left to right. (8) is the auto-probe correlation of the galaxy lensing convergence-tSZ (κ_g-y) combinations, which uses the 10 source redshift bins. (9) is the auto-probe correlation of the CMB lensing convergence-tSZ ($\kappa_{\rm CMB}$-y). (10) is the auto-probe correlation of the tSZ power spectrum (y-y).

Halo Model

Boltzmann Codes

The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological M

Cosmic Shear

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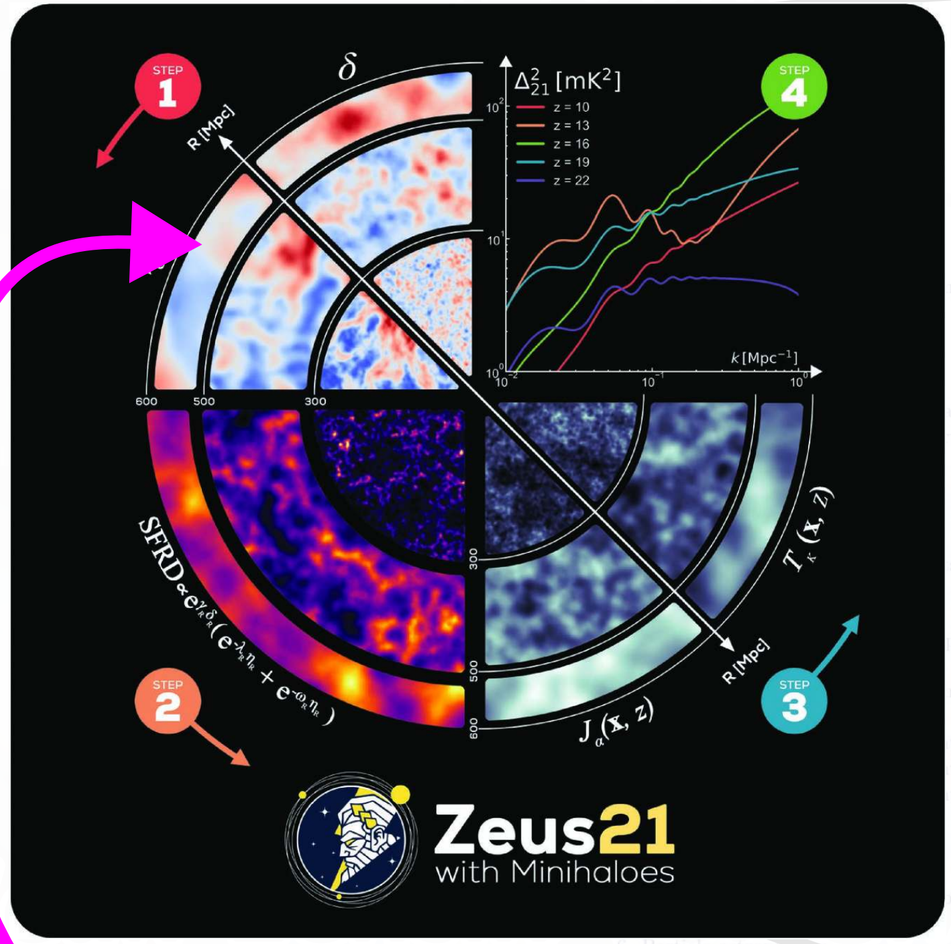
Weak Lensing of CMB

New physics: add a new module to Boltzmann Codes

Cosmic Dawn



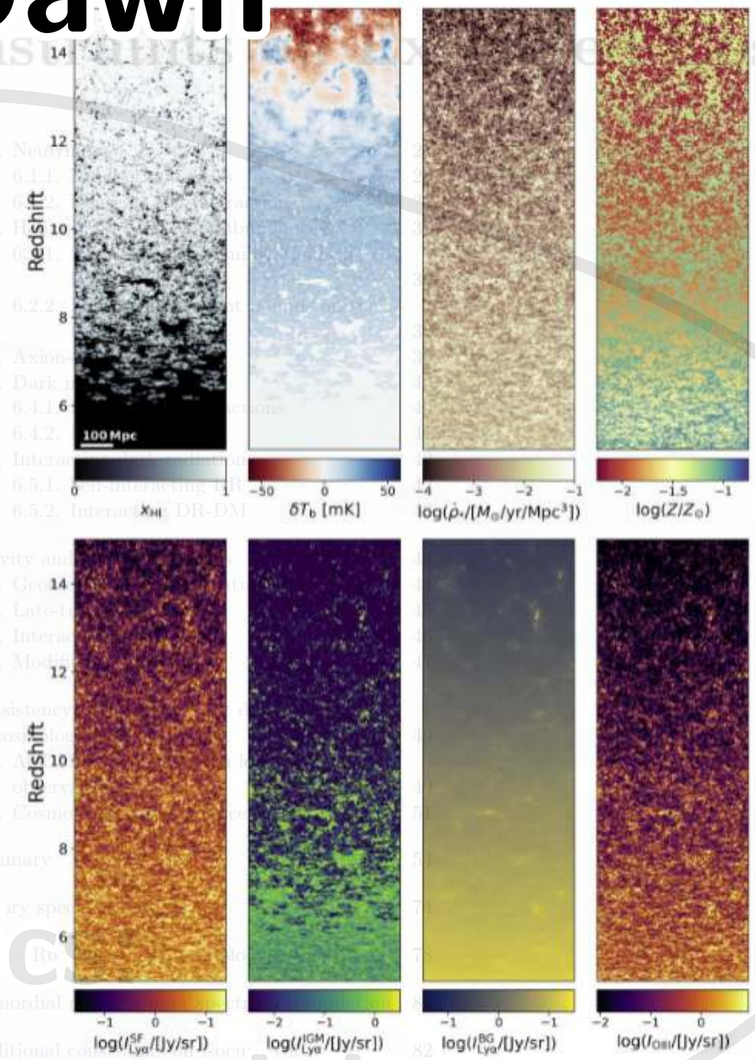
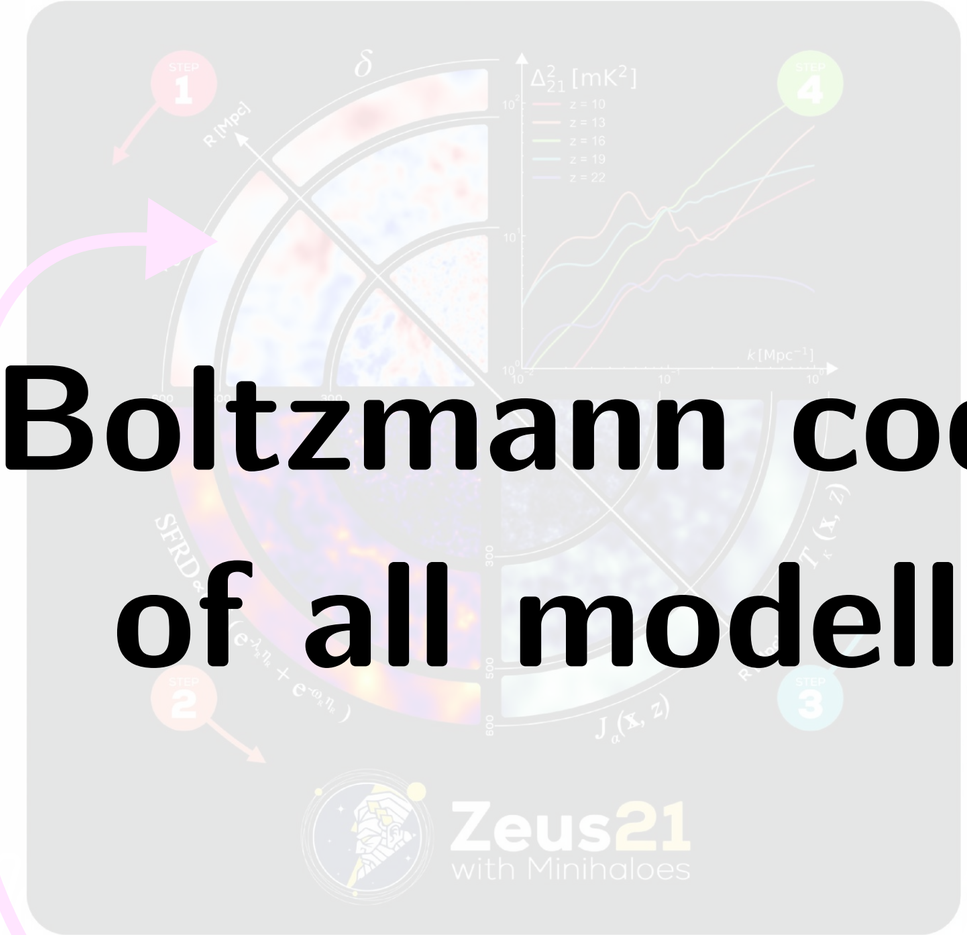


Figure 5. LIMFAST light cones over redshift $5 \leq z \leq 15$. The 8 panels shown correspond to the hydrogen neutral fraction, the brightness temperature of the 21 cm line, the SFRD, the metallicity of collapsed structures, and the intensities of Lyα (from star formation, the ionized IGM, and the scattering background, respectively) and [O III] 5007 Å emission. The color schemes are identically normalized (to the maximum value) for [O III] and Lyα from star formation to highlight their difference.

Boltzmann Codes

Cosmic Dawn



Boltzmann codes the foundation of all modelling in cosmology

Boltzmann Codes

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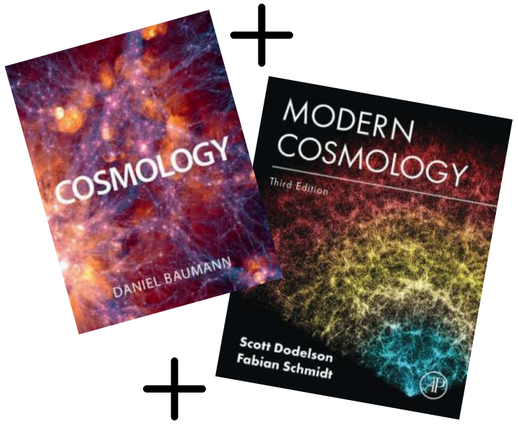

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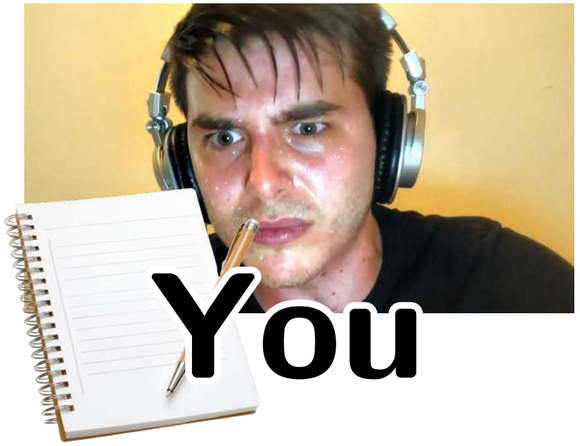

Cosmology course or book

$$
\begin{aligned}
\dot{\Theta} + ik\mu\Theta &= -\dot{\Phi} - ik\mu\Psi \\
&\quad - \dot{\tau}\left[\Theta_0 - \Theta + i\mu v_b - \frac{1}{2}\mathcal{P}_2\Pi\right], \\
\dot{\Theta}_P + ik\mu\Theta_P &= -\dot{\tau}\left[-\Theta_P + \frac{1}{2}(1-\mathcal{P}_2)\Pi\right], \\
\dot{\delta} - kv &= -3\dot{\Phi}, \qquad \dot{v} + \mathcal{H}v = -k\Psi, \\
\dot{\delta}_b - kv_b &= -3\dot{\Phi}, \\
\dot{v}_b + \mathcal{H}v_b &= -k\Psi + \dot{\tau}R\left(v_b + 3\Theta_1\right), \\
\dot{\mathcal{N}} + ik\mu\mathcal{N} &= -\dot{\Phi} - ik\mu\Psi, \\
k^2\Phi + 3\mathcal{H}\left(\dot{\Phi} - \mathcal{H}\Psi\right) &= 4\pi G a^2\left[\rho\delta + \rho_b\delta_b\right. \\
&\qquad \left. + 4\rho_r\Theta_0 + 4\rho_\nu\mathcal{N}_0\right], \\
k^2(\Phi + \Psi) &= -32\pi G a^2\left[\rho_r\Theta_2 + \rho_\nu\mathcal{N}_2\right], \\
\Pi = \Theta_2 + \Theta_2^P + \Theta_0^P, &\qquad R = \frac{4\rho_r}{3\rho_b} = \frac{4\Omega_r}{3\Omega_b a}. \qquad (20)
\end{aligned}
$$

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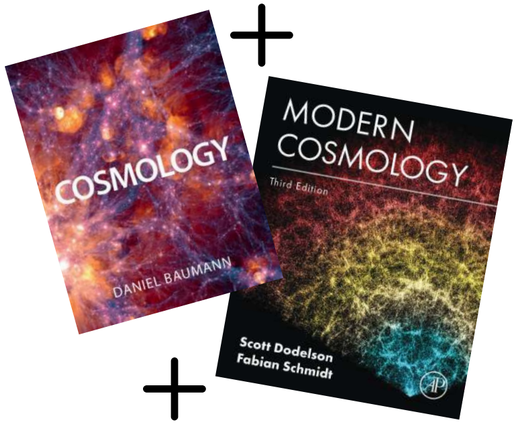

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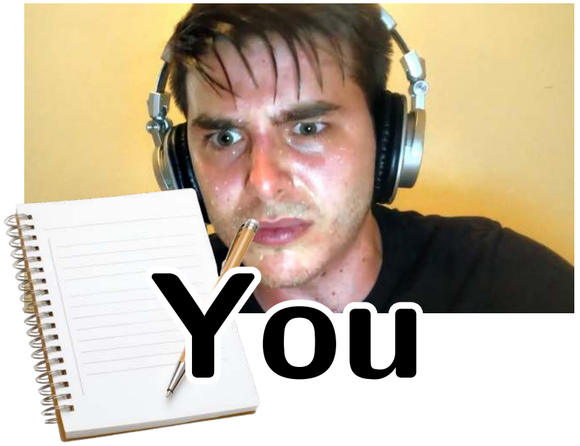

Cosmology course or book

$$
\begin{aligned}
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&\quad - \dot{\tau}\left[\Theta_0 - \Theta + i\mu v_b - \frac{1}{2}\mathcal{P}_2\Pi\right], \\
\dot{\Theta}_P + ik\mu\Theta_P &= -\dot{\tau}\left[-\Theta_P + \frac{1}{2}(1-\mathcal{P}_2)\Pi\right], \\
\dot{\delta} - kv &= -3\dot{\Phi}, \qquad \dot{v} + \mathcal{H}v = -k\Psi, \\
\dot{\delta}_b - kv_b &= -3\dot{\Phi}, \\
\dot{v}_b + \mathcal{H}v_b &= -k\Psi + \dot{\tau}R\left(v_b + 3\Theta_1\right), \\
\dot{\mathcal{N}} + ik\mu\mathcal{N} &= -\dot{\Phi} - ik\mu\Psi, \\
k^2\Phi + 3\mathcal{H}\left(\dot{\Phi} - \mathcal{H}\Psi\right) &= 4\pi G a^2\left[\rho\delta + \rho_b\delta_b\right. \\
&\qquad \left. + 4\rho_r\Theta_0 + 4\rho_\nu\mathcal{N}_0\right], \\
k^2(\Phi + \Psi) &= -32\pi G a^2\left[\rho_r\Theta_2 + \rho_\nu\mathcal{N}_2\right], \\
\Pi = \Theta_2 + \Theta_2^P + \Theta_0^P, &\qquad R = \frac{4\rho_r}{3\rho_b} = \frac{4\Omega_r}{3\Omega_b a}. \qquad (20)
\end{aligned}
$$

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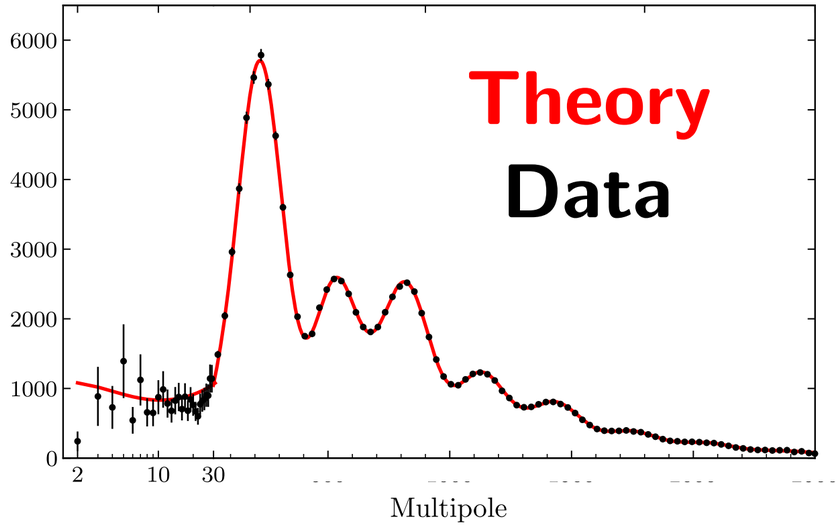

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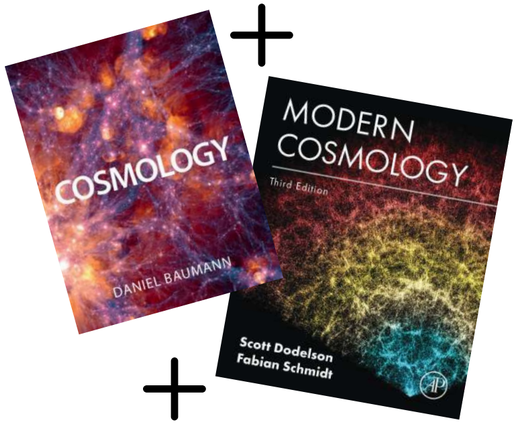

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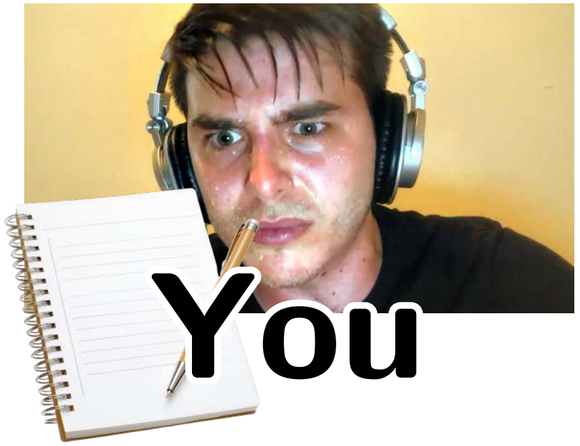

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$$\dot{\Theta}_P + ik\mu\Theta_P = -\dot{\tau}\left[-\Theta_P + \frac{1}{2}(1-\mathcal{P}_2)\Pi\right]$$

$$\dot{\delta} -$$

$$\dot{\delta}_b - k$$

$$\dot{v}_b + \mathcal{H}$$

$$\dot{\mathcal{N}} + ik\mu$$

$$k^2\Phi + 3\mathcal{H}\left(\dot{\Phi} - \mathcal{H}\Psi\right.$$

$$k^2(\Phi + \Psi) = -32\pi Ga^2\left[\rho_r\Theta_2 + \rho_\nu\mathcal{N}_2\right],$$

$$\Pi = \Theta_2 + \Theta_2^P + \Theta_0^P, \qquad R = \frac{4\rho_r}{3\rho_b} = \frac{4\Omega_r}{3\Omega_b a}. \qquad (20)$$

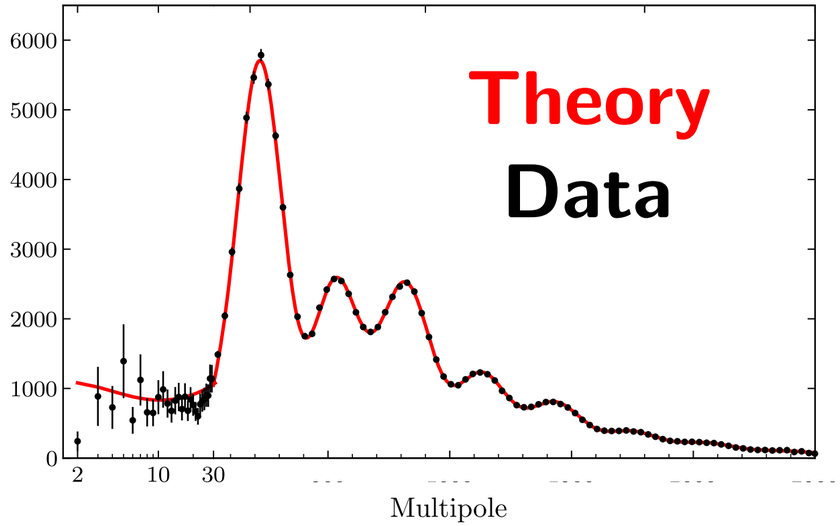
7.7 CLASS

Download and install the CLASS software:

https://lesgourg.github.io/class_public/class.html

Use it to reproduce some of the figures in this chapter.

Baumann Problem 7.7



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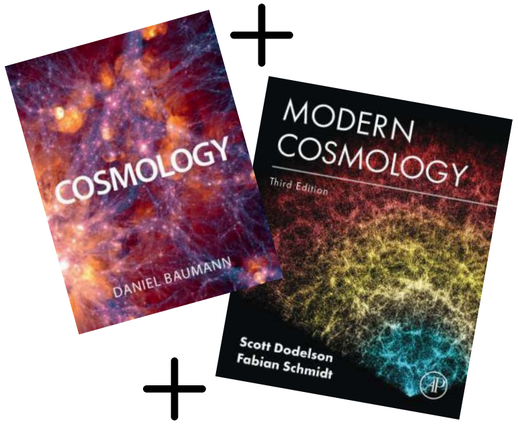

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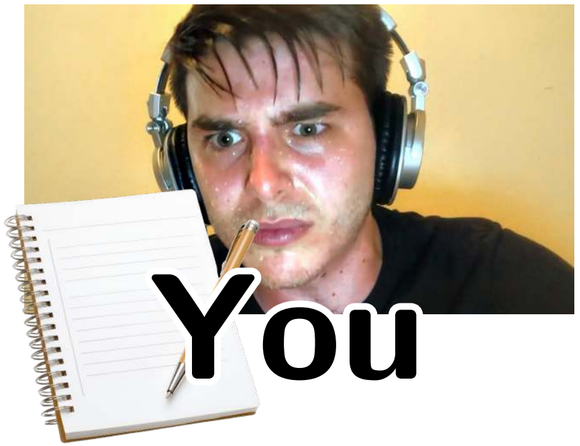

Cosmology course or book

(can't get out of their own way)

$$\dot{\delta}_b - k$$
$$\dot{v}_b + \mathcal{H}$$
$$\dot{\mathcal{N}} + ik\mu$$
$$k^2\Phi + 3\mathcal{H}\left(\dot{\Phi} - \mathcal{H}\Psi\right.$$

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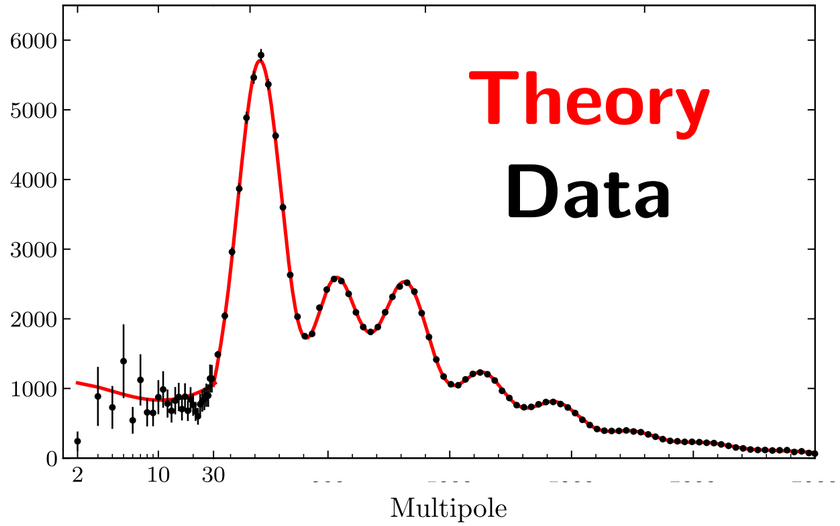
$$\Pi = \Theta_2 + \Theta_2^P + \Theta_0^P, \qquad R = \frac{4\rho_r}{3\rho_b} = \frac{4\Omega_r}{3\Omega_b a}. \qquad (20)$$



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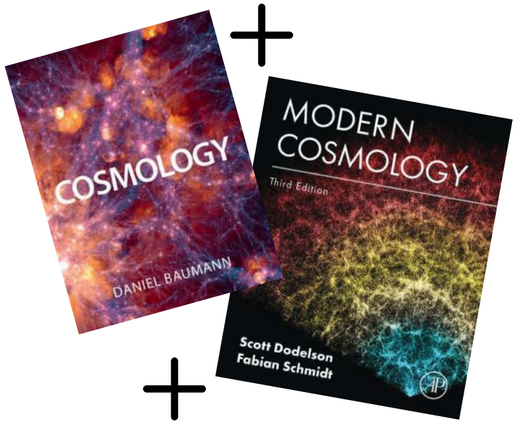

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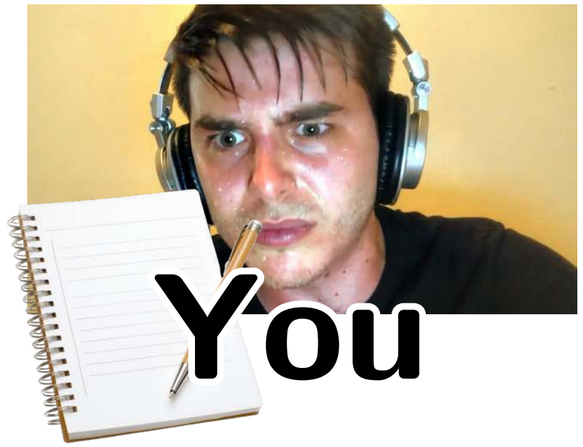


Cosmology course or book

(can't get out of their own way)

$$+ \dot{\Theta} \ldots \mu \ldots$$

$$- \dot{\tau} \left[\Theta_0 - \Theta + i\mu v_b - \frac{1}{2}\mathcal{P}_2 \Pi \right],$$

$$\Theta_P + ik\mu\Theta_P = \ldots \left[-\Theta_P + \ldots \right]$$

$$\dot{\delta} - \ldots$$

$$\dot{\delta}_b - k \ldots$$

$$\dot{v}_b + \mathcal{H} \ldots$$

$$\dot{\mathcal{N}} + ik\mu \ldots$$

$$k^2\Phi + 3\mathcal{H}\left(\dot{\Phi} - \mathcal{H}\Psi \ldots\right.$$

$$k^2(\Phi + \Psi) = -32\pi G a^2 \left[\rho_r \Theta_2 + \rho_\nu \mathcal{N}_2 \right],$$

$$\Pi = \Theta_2 + \Theta_2^P + \Theta_0^P, \qquad R = \frac{4\rho_r}{3\rho_b} = \frac{4\Omega_r}{3\Omega_b a}. \qquad (20)$$

7.7 CLASS

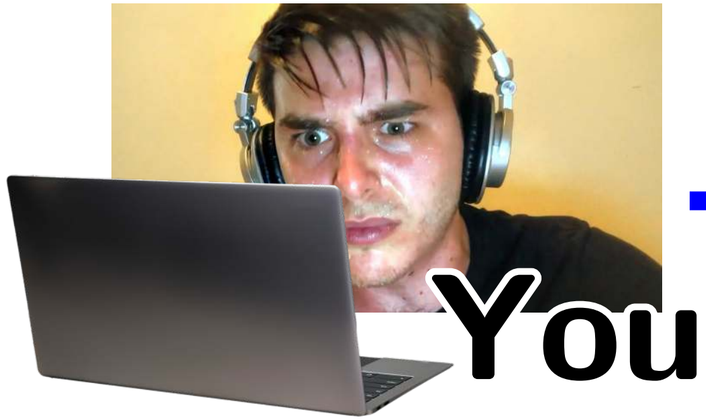
Download and install the CLASS software:

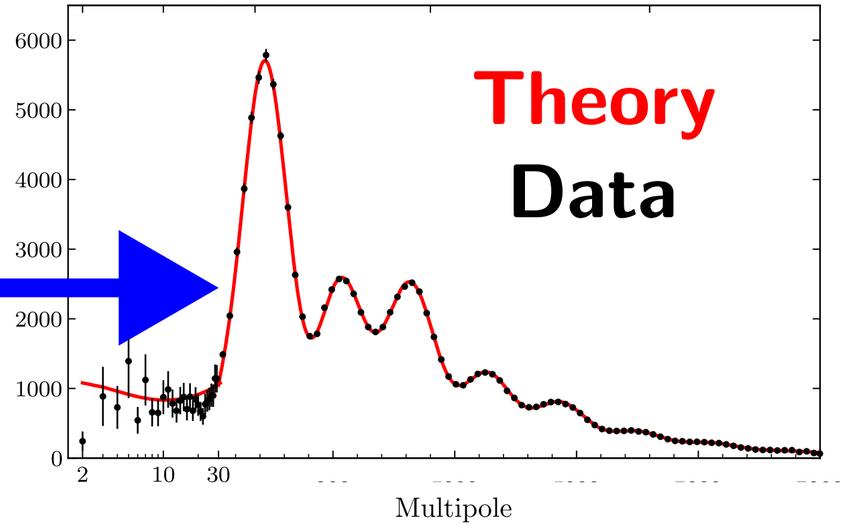
https://lesgourg.github.io/class_public/class.html

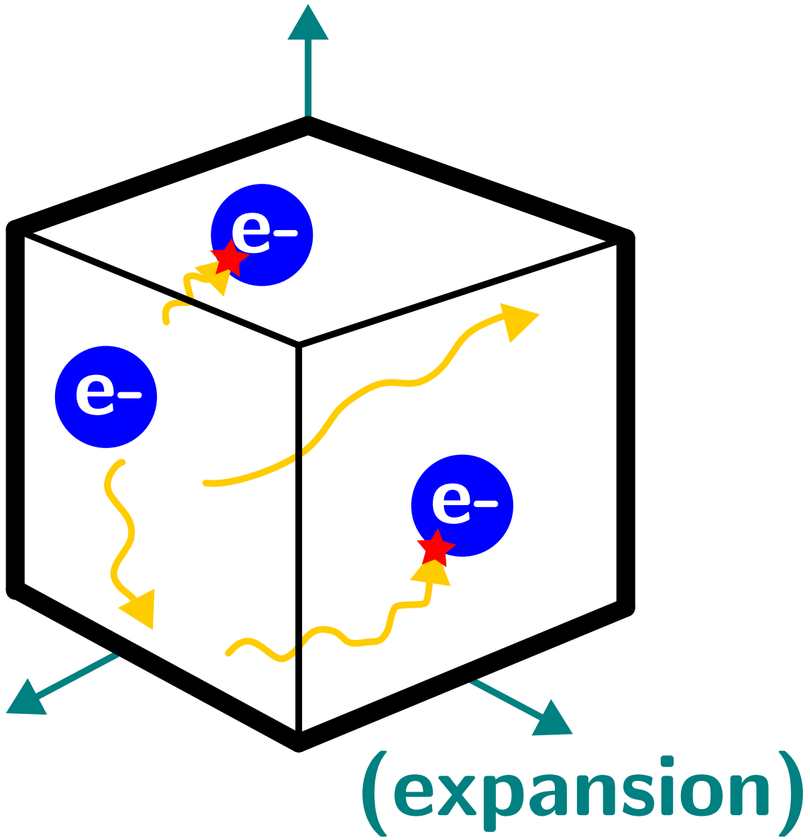
Use it to reproduce some of the figures in this chapter.

Baumann Problem 7.7

This Section



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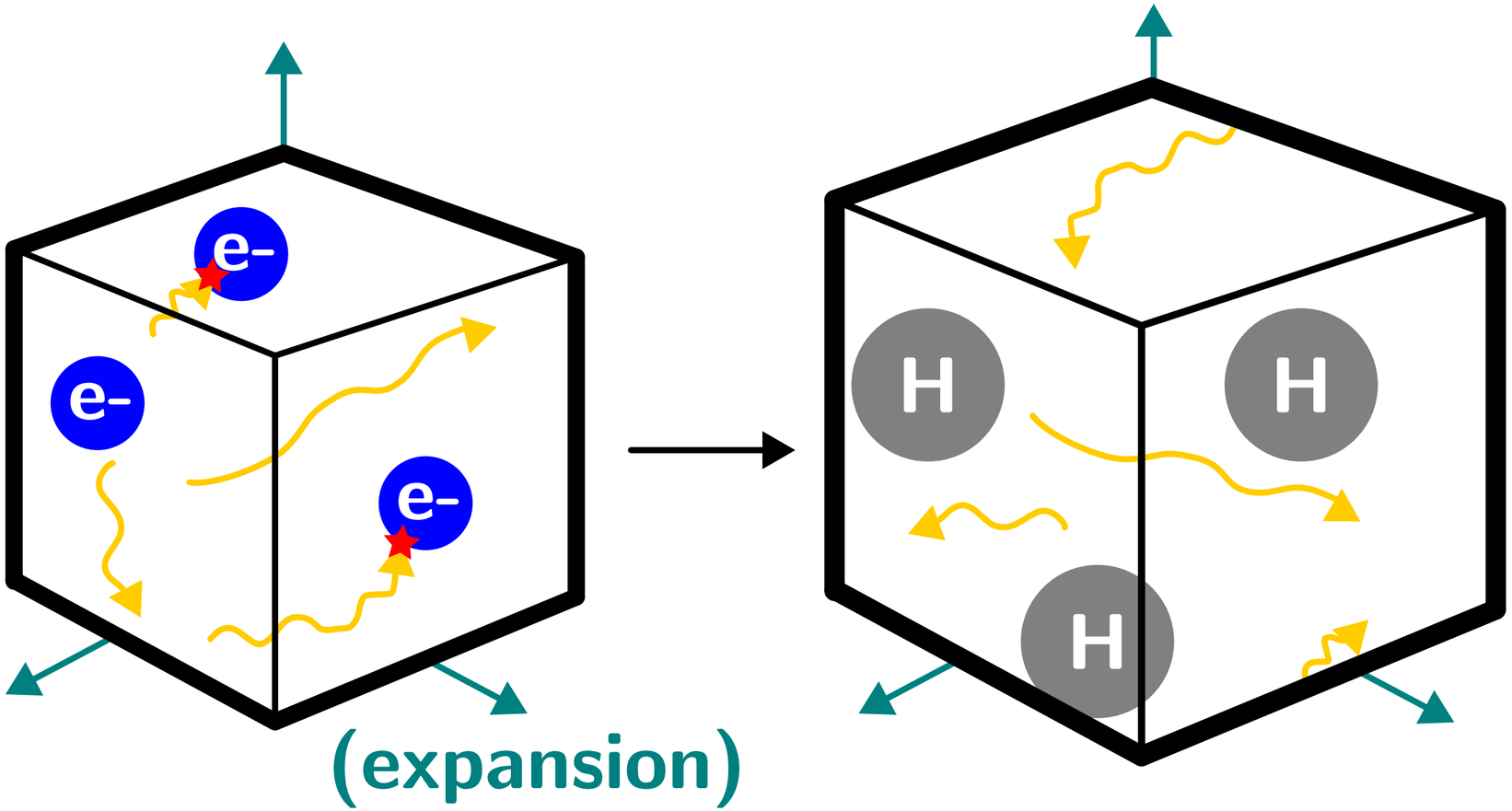
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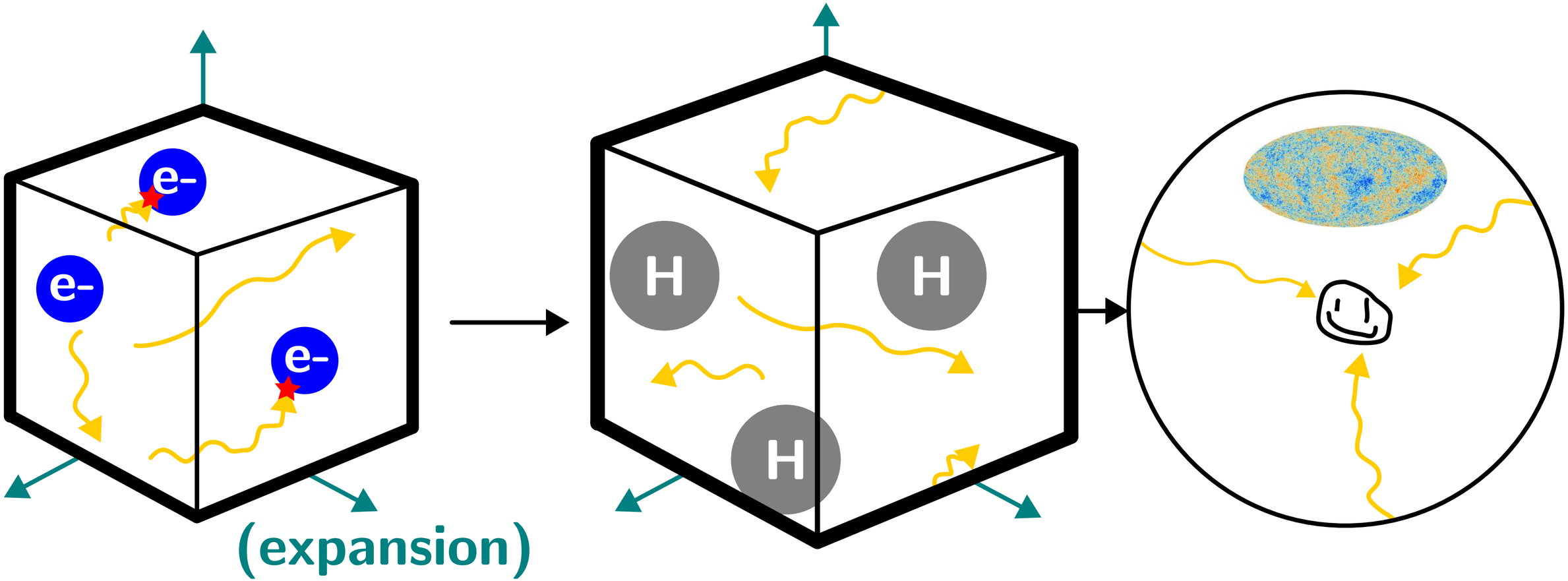
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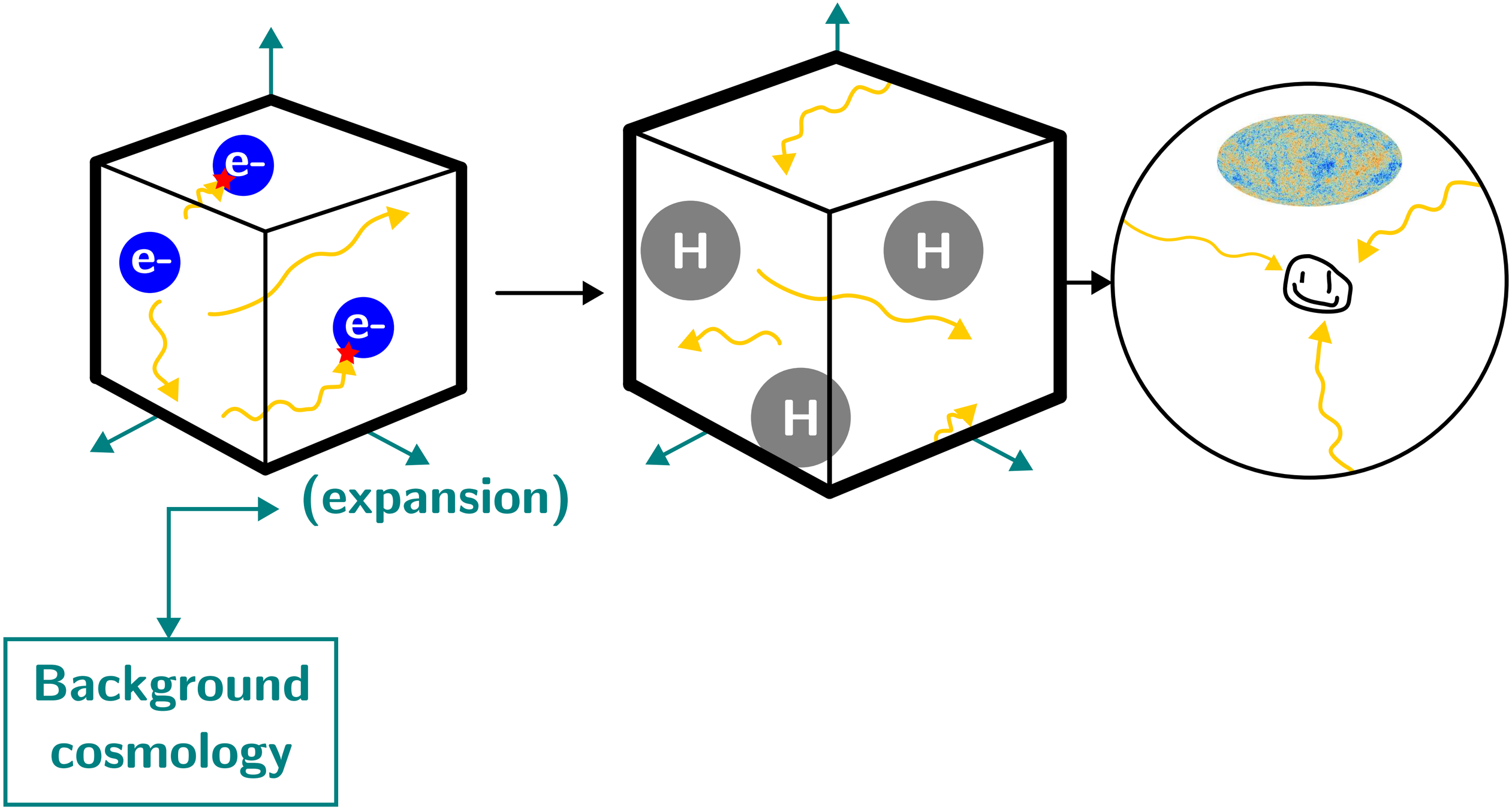
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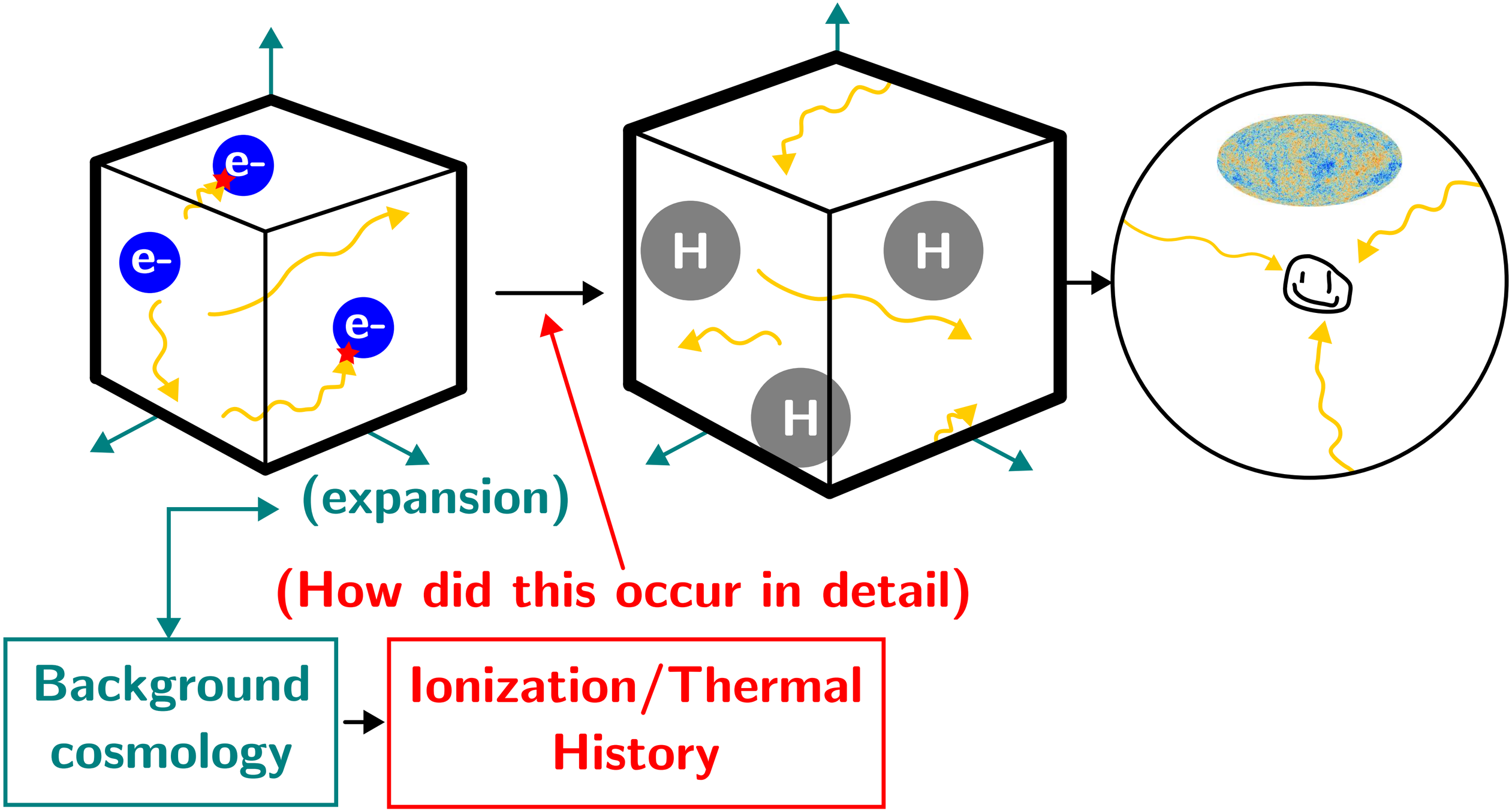
Background cosmology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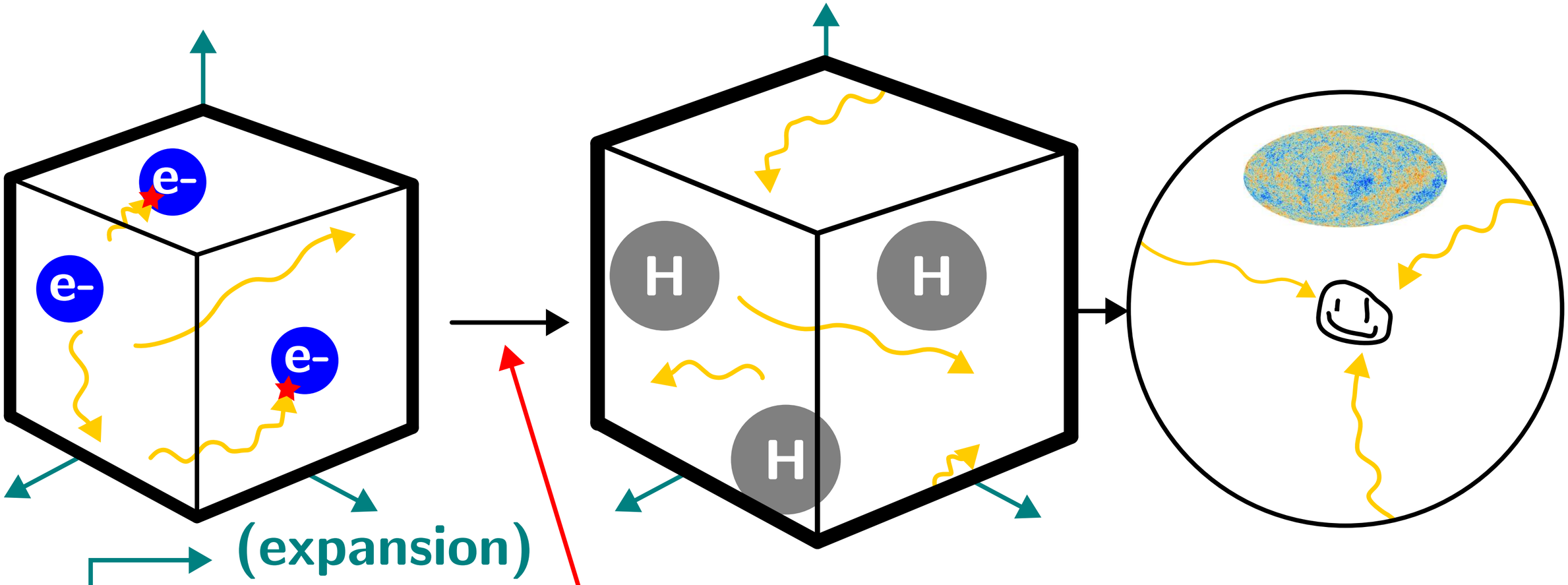
(How did this occur in detail)
Background cosmology
Ionization/Thermal History



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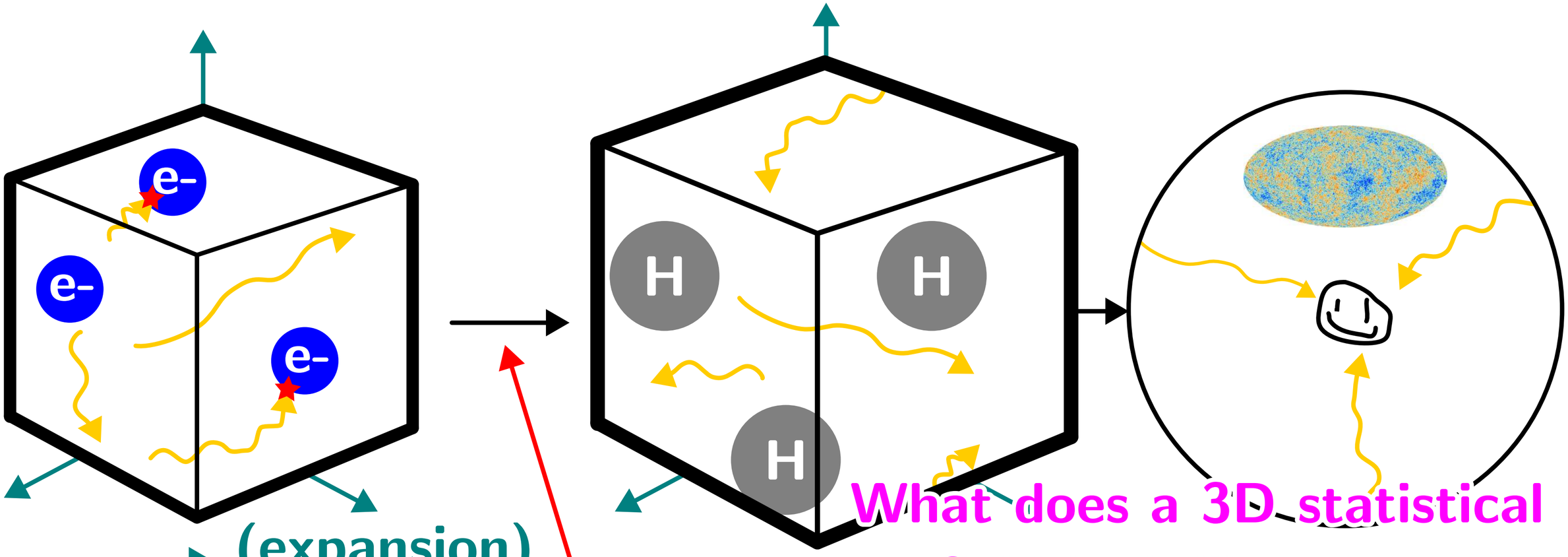
(How did this occur in detail)

Background cosmology → Ionization/Thermal History → Perturbation Evolution

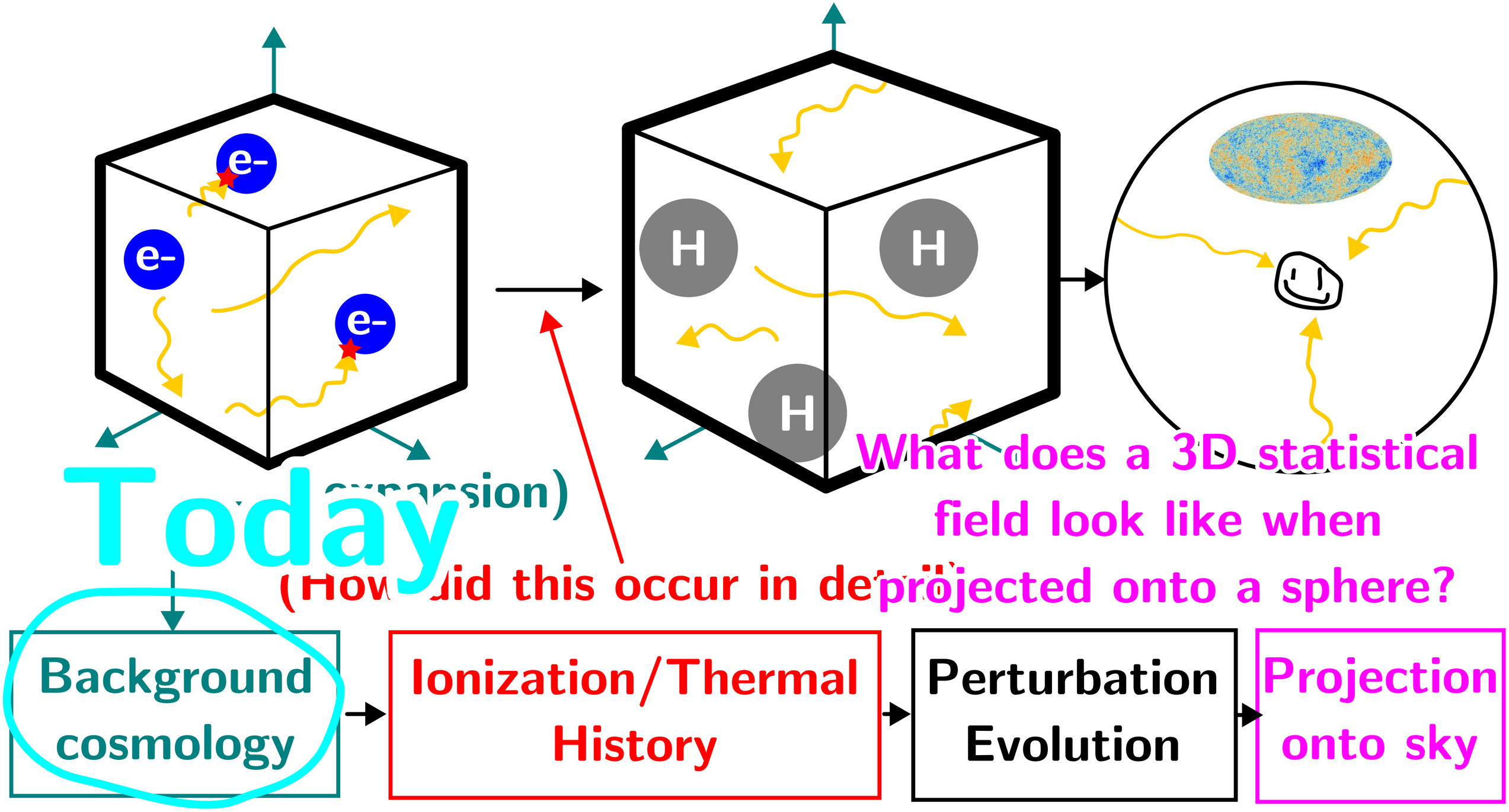


(expansion)

What does a 3D statistical field look like when projected onto a sphere?

(How did this occur in detail?)

Background cosmology → Ionization/Thermal History → Perturbation Evolution → Projection onto sky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expansion)
What does a 3D statistical field look like when projected onto a sphere?
(How did this occur in detail)
Background cosmology
Ionization/Thermal History
Perturbation Evolution
Projection onto sky